

OPTIMAL SCORING STRATEGIES FOR THE LIZARD CULT

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We propose a discrete-time Markov chain framework to model scoring using gardens for the Lizard Cult faction in the asymmetric board game Root. The Cult's ability to score from their gardens is affected by three different processes with stochasticity introduced by their opponents: suited card draws from a shared deck, the dynamic Outcast suit determined by recently discarded cards, and forcible removal of their gardens. Unlike other factions, the Cult chooses when it will score from its gardens, which allows for significant strategy-based variation. We use Monte Carlo simulation to visualize scoring trajectories and determine which parameters are most key.

1 INTRODUCTION

Root is a board game with a high degree of asymmetry: each player chooses a unique faction out of 10. Every faction has unique actions and their own ways to score and 'win the war'. For the Lizard Cult faction, the primary victory mechanism involves building *gardens* and performing the Score ritual "citeroot" (Appendix F). Most other factions have automatic or incidental automatic scoring, but the Cult chooses when it scores points (at the expense of its resources), which creates an optimization problem. Three sources of randomness complicate this optimization:

1. **Card draws:** Rituals require suited cards drawn from a shared 54-card deck. The Cult's 'Hatred of Birds' rule means the 18 Bird-suit cards cannot be used for scoring.
2. **Garden dynamics:** Gardens grow in number through player actions (the Build ritual, the Cult's Conspiracies) but are likely to be destroyed by opponents through combat and special abilities.
3. **Hand composition:** The random distribution of which suits of cards will be drawn cards constrains the suits which can be scored each turn.

1.1 Game Mechanics

The Cult's faction board has three garden tracks (Fox, Rabbit, Mouse), each with capacity 5. Let $\mathbf{G} = (G_F, G_R, G_M) \in \{0, \dots, 5\}^3$ denote gardens on the map. The Cult starts with 1 garden (of chosen suit) on the map "citeroot".

Scoring. On the Cult's turn, performing the Score ritual for suit s yields victory points based on garden

count, as per the faction board:

$$S(g) = \begin{cases} 0 & g \in \{0, 1\} \\ 2 & g \in \{2, 3\} \\ 3 & g = 4 \\ 4 & g = 5 \end{cases} \quad (1)$$

Card draw. The Cult draws $n(\mathbf{G}) = 1 + \text{Bonus}(\mathbf{G})$ cards per turn, where

$$\text{Bonus}(\mathbf{G}) = \sum_s \mathbb{1}[G_s \geq 2], \quad (2)$$

counts draw bonuses (one per suit with ≥ 2 gardens). After drawing, the Cult must discard down to a hand limit of 5 cards. Since discards are chosen by the player, this allows preferential retention of suited cards for scoring on subsequent turns.

Scoring constraints. Each suit can be scored at most once per turn, but the Cult can score multiple suits (up to 3) in a single turn. Scoring requires spending a suited card.

Victory. The first player to 30 VP wins immediately.

1.2 Threshold Policies

We formalize the Cult's choice as to whether they should score or wait as a threshold policy.

Definition (Threshold Policy). Our threshold policy π_θ with parameter $\theta \in \{1, \dots, 5\}$ is:

- **Score** suit s if $G_s \geq \theta$ and a suit- s card is available
- **Build**¹ otherwise

Definition (Victory Time). The victory time under policy π_θ is the hitting time

$$\tau(\theta) = \inf\{t \geq 0 : V_t \geq 30\} \quad (3)$$

and we define $\theta^* = \arg \min_\theta \mathbb{E}[\tau(\theta)]$, the minimizer of expected turns to victory.

¹The mechanisms that 'Build' encompasses are vast, and we cannot model them all here. In general, the Cult should use the Sacrifice ritual on Bird cards, and complete Recruit rituals. These will complement the Build ritual and Sanctify conspiracy, which allow the Cult to get more gardens on the map.

Standard Root Map (Fall)

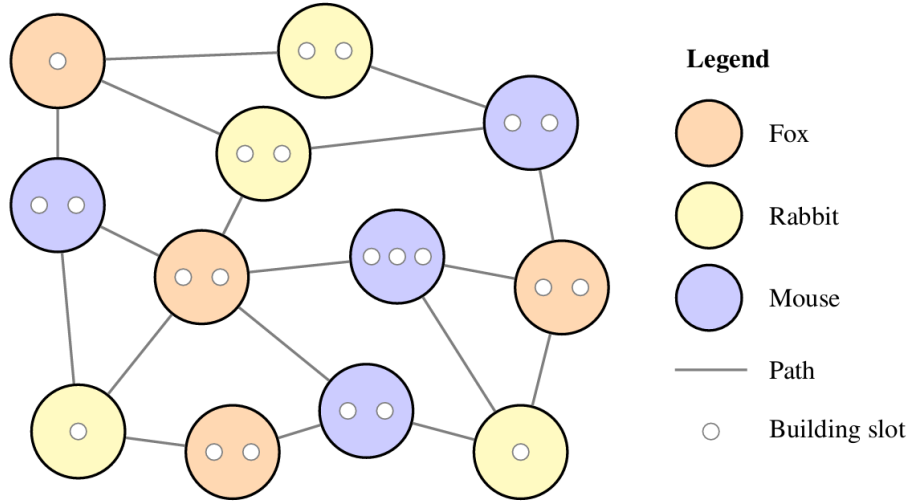


Figure 1: The standard (Fall) map. Clearings are coloured by suit; white circles indicate all possible building slots where the Lizard Cult can place gardens (note: in some games, certain slots have Ruins and are unusable) [1].

2 CARD DRAW ANALYSIS

2.1 The Hypergeometric Distribution

The deck contains 54 cards: 12 Fox, 12 Rabbit, 12 Mouse, 18 Bird. The Cult's 'Hatred of Birds' means only 36 cards (two-thirds) are usable for scoring.

Proposition 1 (Draw Distribution). *Drawing n cards, the number of suit- s cards follows*

$$X_s \sim \text{Hypergeometric}(54, 12, n) \quad (4)$$

with mean $\mathbb{E}[X_s] = 12n/54 = 2n/9$.

Proof. This is a standard hypergeometric distribution with parameters making it mathematically equivalent to drawing cards from a deck without replacement [4]. The mean follows from linearity of expectation. $\square$

Proposition 2 (Hit Probability). *The probability of drawing at least one card of suit s is*

$$q(n) = \mathbb{P}(X_s \geq 1) = 1 - \frac{\binom{42}{n}}{\binom{54}{n}}. \quad (5)$$

Since the Cult can score multiple suits per turn, the expected VP per turn depends not just on the per-suit hit probability $q(n)$ but on how many distinct suits appear in a single draw. We therefore consider $K = |\{s : X_s \geq 1\}|$, the number of scorable suits (i.e., those with at least one available card).

Cards drawn n	1	2	3	4
$q(n)$	0.222	0.399	0.540	0.652
$\mathbb{E}[\text{suited cards}]$	0.67	1.33	2.00	2.67
$\mathbb{E}[K]$	0.67	1.20	1.62	2.20

Furthermore, for $n = 4$ draws from a fresh deck:

$K \geq$	1	2	3
Probability	0.990	0.759	0.418

In particular, with full draw bonuses ($n = 4$), the Cult has 76% chance of scoring 2+ suits per turn.

Proof. See (Appendix B). $\square$

3 MARKOV CHAIN MODEL

We define a finite-state Markov chain that captures the essence of the relevant strategic working of the Cult.

3.1 Reduced State Space

Tracking the full game state is both overly complicated and quite difficult; the state space becomes intractably large. We instead define a reduced state space for the key aspects of strategy.

Definition (Reduced State Space). *The state on turn t for the state space $\mathcal{S}$ is given by*

$$\mathbf{X}_t = (\mathbf{G}_t, V_t, \mathbf{h}_t) \in \mathcal{S} \quad (6)$$

where:

- $\mathbf{G}_t = (G_F^t, G_R^t, G_M^t) \in \{0, 1, 2, 3, 4, 5\}^3$: garden counts per suit
- $V_t \in \{0, 1, \dots, 30\}$: victory points (states with $V \geq 30$ are absorbing)
- $\mathbf{h}_t = (h_F^t, h_R^t, h_M^t) \in \{0, 1\}^3$: indicators where $h_s = 1$ iff at least one suit- s card is in hand.

The garden count is given by $|\mathbf{G}| = G_F + G_R + G_M$. The state space has $|\mathcal{S}| = 6^3 \times 30 \times 2^3 = 51,840$ transient states, and one absorbing ‘victory’ state.

The hand indicator $\mathbf{h}$ captures whether scoring suit s is possible this turn, instead of considering how many suited cards are available. This does not affect the analysis, as the Cult can only score each suit once per turn. The other approximation is a fresh deck: in the model, we draw from the full 54-card deck each turn. In particular, draws from the deck are independent of existing hand composition. This simplification is necessary for computation (otherwise, there are 41,743 deck states) and reasonable since the Cult only cares about suit distribution (and not specific cards).

3.2 Turn Structure and Transition Kernel

We adapt *Root*’s turn structure language of *Birdsong*, *Daylight*, and *Evening*. In *Birdsong*, the Cult can Sanctify new gardens into play; in *Daylight*, they can perform rituals, including Build and Score. Logically, we assume that players will Build before they Score, allowing us to simplify. They then Draw, and pass to other players. We model this in four phases: growth, scoring, card draw, then attrition.

$$(\mathbf{G}, V, \mathbf{h}) \xrightarrow{1: \text{Growth}} \mathbf{G}' \xrightarrow{2: \text{Score}} (V', \mathbf{h}') \xrightarrow{3: \text{Draw}} \mathbf{h}'' \xrightarrow{4: \text{Attrition}} (\mathbf{G}'', \mathbf{h}''').$$

The next-turn state is $\mathbf{X}_{t+1} = (\mathbf{G}'', V', \mathbf{h}''')$.

3.2.1 Phase 1: Growth of Gardens

Gardens can grow through Build rituals and the Sanctify conspiracy; since this depends highly on player behaviour, luck, and other factors, we choose to model this as independent Bernoulli growth using an aggregate parameter μ .

Definition (Growth Transition). *Each suit’s garden count increases by 1 with probability μ , capped at 5:*

$$\begin{aligned} \mathbb{P}(G'_s = g+1 \mid G_s = g) &= \mu \cdot \mathbb{1}[g < 5] \\ \mathbb{P}(G'_s = g \mid G_s = g) &= 1 - \mu \cdot \mathbb{1}[g < 5] \end{aligned} \quad (7)$$

Suits grow independently, so

$$\mathbb{P}(\mathbf{G}' \mid \mathbf{G}) = \prod_{s \in \{F, R, M\}} \mathbb{P}(G'_s \mid G_s). \quad (8)$$

Remark (Interpretation of μ). In practice, $\mu \in [0.2, 0.5]$ depending on luck and opponent pressure (for example, the Cult may need to defend gardens instead of building new ones).

3.2.2 Phase 2: Scoring

Under a threshold policy π_θ , the Cult scores all eligible suits.

Definition (Scoring Transition). *Given threshold θ , garden configuration $\mathbf{G}'$, and hand $\mathbf{h}$:*

- For each suit s : if $G'_s \geq \theta$ and $h_s = 1$, score $S(G'_s)$ points and set $h'_s = 0$.
- Otherwise: $h'_s = h_s$.

The VP update is

$$V' = \min \left(V + \sum_s S(G'_s) \cdot \mathbb{1}[G'_s \geq \theta] \cdot h'_s, 30 \right). \quad (9)$$

The hand update is

$$h'_s = h_s \cdot (1 - \mathbb{1}[G'_s \geq \theta]). \quad (10)$$

3.2.3 Phase 3: Card Draw and Hand Update

The Cult draws cards based on draw bonuses, then updates hand composition.

Definition (Draw Count). *Given garden configuration $\mathbf{G}'$, the number of cards drawn is*

$$n(\mathbf{G}') = 1 + \text{Bonus}(\mathbf{G}') = 1 + \sum_s \mathbb{1}[G'_s \geq 2]. \quad (11)$$

Definition (Hand Transition). *Let $\mathbf{Y} = (Y_F, Y_R, Y_M, Y_B)$ be the cards drawn, where Y_s is the count of suit- s cards, and denote $n = n(\mathbf{G}')$. Under a fresh-deck approximation, $\mathbf{Y}$ follows a multivariate hypergeometric distribution:*

$$\mathbf{Y} \sim \text{MultiHypergeom}(54; 12, 12, 12, 18; n). \quad (12)$$

After optimal discard, the new hand indicator is

$$h''_s = \mathbb{1}[h_s = 1 \text{ or } Y_s \geq 1]. \quad (13)$$

Since we only need to know whether $Y_s \geq 1$, the transition probability simplifies to

$$\mathbb{P}(h''_s = 1 \mid h_s, n) = \begin{cases} 1 & \text{if } h_s = 1 \\ q(n) & \text{if } h_s = 0 \end{cases} \quad (14)$$

where $q(n)$ is the probability of drawing at least one card of suit s (see Proposition 2).

Due to the fresh-deck approximation, drawing the suits are conditionally independent:

$$\mathbb{P}(\mathbf{h}'' \mid \mathbf{h}', \mathbf{G}') = \prod_s \mathbb{P}(h''_s \mid h'_s, n(\mathbf{G}')). \quad (15)$$

3.2.4 Phase 4: Attrition

Opponents destroy gardens through combat and special abilities; again, as this depends so heavily on factors outside of player control, we introduce a parameter to simulate interference. Also, each garden destroyed forces the Cult to discard one card at random from their hand.

Definition (Attrition Model). *Total garden losses in a turn follow $L \sim \text{Poisson}(\lambda)$, where $\lambda > 0$ is the interference rate. We assume that losses are allocated uniformly at random among existing gardens. Given total losses $L = \ell$ and garden configuration $\mathbf{G}'$ with $|\mathbf{G}'| > 0$, the per-suit losses (L_F, L_R, L_M) follow*

$$(L_F, L_R, L_M) \mid L = \ell, \mathbf{G}' \sim \text{Mltnom} \left(\ell; \frac{G'_F}{|\mathbf{G}'|}, \frac{G'_R}{|\mathbf{G}'|}, \frac{G'_M}{|\mathbf{G}'|} \right)$$

where Mltnom represents the Multinomial distribution. The updated garden counts are

$$G''_s = \max(G'_s - L_s, 0). \quad (16)$$

Assumption (Random Forced Discard) Each destroyed garden forces the Cult to discard a card randomly (after the Cult has already discarded down to no more than the hand limit $H = 5$).

With $D = \sum_s L_s$ gardens destroyed, the Cult discards D cards sequentially from a shrinking hand of sizes $H, H-1, \dots$ which is equivalent to choosing D cards uniformly without replacement from H cards.

In the binary hand model, suit s is eliminated from hand iff all c_s copies of suit- s cards are among the D discards:

$$\mathbb{P}(\text{eliminate } s \mid c_s, H, D) = \frac{\binom{H-c_s}{D-c_s}}{\binom{H}{D}} \cdot \mathbb{1}[D \geq c_s]. \quad (17)$$

Since c_s is unobserved in the binary model, we average over the conditional distribution $c_s \mid c_s \geq 1$. Under the fresh-deck approximation, the hand of $H = 5$ is treated as a random sample from the 54-card deck, so $c_s \sim \text{Hypergeometric}(54, 12, 5)$. Let $\omega_c = \mathbb{P}(c_s = c \mid c_s \geq 1)$ be the conditional weights:

c	1	2	3	≥ 4
ω_c	0.581	0.328	0.082	0.009

The suit elimination probability for D discards is then

$$\gamma(D) = \mathbb{P}(h'''_s = 0 \mid h''_s = 1, D) = \sum_{c=1}^{\min(H,D)} \omega_c \frac{\binom{H-c}{D-c}}{\binom{H}{D}}$$

with the post-attrition hand survival

$$\mathbb{P}(h'''_s = 1 \mid h''_s = 1, D) = 1 - \gamma(D) \quad (18)$$

and $\mathbb{P}(h'''_s = 1 \mid h''_s = 0) = 0$.

Precomputed values of γ for $H = 5$:

Gardens lost D	1	2	3
$\gamma(D)$ (elim. prob.)	0.116	0.265	0.455

At typical attrition rates ($\lambda = 0.2$), $\mathbb{P}(D \geq 1) \approx 0.18$ and $\mathbb{P}(D \geq 2) \approx 0.02$, so forced discards affect hand composition only occasionally.

Remark (Interpretation of the rate λ). The parameter λ is the expected number of gardens destroyed per turn by opponents. In passive games, $\lambda \approx 0.1$ – 0.2 ; in aggressive games where opponents target the Cult, $\lambda \approx 0.4$ – 0.6 . In most games, λ might vary throughout the game, and soar if the Cult is in a position to win.

Proposition 3 (Attrition Transition Probabilities). *For a single suit with g gardens (out of $|\mathbf{G}|$ total), the probability of losing exactly k gardens is given by the Poisson-Multinomial compound*

$$\mathbb{P}(L_s = k \mid \mathbf{G}') = \sum_{\ell=k}^{\infty} \frac{e^{-\lambda} \lambda^\ell}{\ell!} \binom{\ell}{k} \left(\frac{g}{|\mathbf{G}'|} \right)^k \left(1 - \frac{g}{|\mathbf{G}'|} \right)^{\ell-k}$$

(for computational purposes, we truncate at $\ell = 10$ since $\mathbb{P}(L > 10) \approx 0$).

In particular, the probability of losing zero gardens in suit s ('survival probability') is

$$\mathbb{P}(L_s = 0 \mid \mathbf{G}') = e^{-\lambda g/|\mathbf{G}'|} = p_{\text{surv}}. \quad (19)$$

Proof. By the Poisson decomposition property, if $L \sim \text{Poisson}(\lambda)$ and each loss hits suit s independently with probability $p = g/|\mathbf{G}'|$, then $L_s \sim \text{Poisson}(\lambda p)$, which implies $\mathbb{P}(L_s = 0) = e^{-\lambda p}$; see Ross [2], Chapter 5, Section 5.3. [2]. $\square$

3.3 Full Transition Kernel

Definition (Transition Kernel). *Under threshold policy π_θ , the transition probability from state $\mathbf{x} = (\mathbf{G}, V, \mathbf{h})$ to state $\mathbf{x}' = (\mathbf{G}', V', \mathbf{h}''')$ is*

$$\mathbb{P}_\theta(\mathbf{x}' \mid \mathbf{x}) = \sum_{\mathbf{G}', \mathbf{h}', \mathbf{h}''} \mathbb{P}_{\text{grow}}(\mathbf{G}' \mid \mathbf{G}) \cdot \mathbb{P}_{\text{draw}}(\mathbf{h}'' \mid \mathbf{h}', \mathbf{G}') \cdot \mathbb{1}[\text{score}(\mathbf{G}', \mathbf{h}, V; \theta) = (V', \mathbf{h}')] \cdot \mathbb{P}_{\text{attrit}}(\mathbf{G}'', \mathbf{h}''' \mid \mathbf{G}', \mathbf{h}'')$$

where $\text{score}(\cdot)$ is the deterministic scoring function from Definition 3.2.2, and $\mathbb{P}_{\text{attrit}}$ includes the random forced discard from Assumption 3.2.4:

$$\mathbb{P}_{\text{attrit}}(\mathbf{G}'', \mathbf{h}''' \mid \mathbf{G}', \mathbf{h}'') = \mathbb{P}_{\text{attrit}}(\mathbf{G}'' \mid \mathbf{G}') \cdot \prod_s \mathbb{P}(h'''_s \mid h''_s, D)$$

with $D = \sum_s (G'_s - G''_s)$ and per-suit hand survival given above.

Definition (Transition Matrix). We construct the transition matrix $\mathbf{P}_\theta$, a $(|\mathcal{S}| + 1) \times (|\mathcal{S}| + 1)$ matrix with entries $[\mathbf{P}_\theta]_{\mathbf{x}, \mathbf{x}'} = \mathbb{P}_\theta(\mathbf{x}' | \mathbf{x})$. States with $V' \geq 30$ transition to the absorbing victory state with probability 1. Let

$$\mathbf{P}_\theta = \begin{pmatrix} \mathbf{Q}_\theta & \mathbf{r}_\theta \\ \mathbf{0}^\top & 1 \end{pmatrix} \quad (20)$$

where $\mathbf{Q}_\theta$ is the $|\mathcal{S}| \times |\mathcal{S}|$ submatrix of transient-to-transient transitions, $\mathbf{0}$ is the zero vector of dimension $|\mathcal{S}|$, and $\mathbf{r}_\theta \in [0, 1]^{|\mathcal{S}|}$ is the absorption vector whose $\mathbf{x}$ -th entry gives the probability of transitioning from transient state $\mathbf{x}$ directly to the absorbing victory state in one step:

$$[\mathbf{r}_\theta]_{\mathbf{x}} = \mathbb{P}_\theta(V' \geq 30 | \mathbf{X}_t = \mathbf{x}). \quad (21)$$

Note that absorption can only occur during the scoring phase (Phase 2); the subsequent draw and attrition phases cannot increase VP.

3.4 Victory Time Analysis

Definition (Victory Time). The victory time under policy π_θ starting from state $\mathbf{x}$ is

$$\tau_\theta(\mathbf{x}) = \inf\{t \geq 0 : V_t \geq 30 | \mathbf{X}_0 = \mathbf{x}\}. \quad (22)$$

Proposition 4 (Hitting Time Moments). Let $\mathbf{N}_\theta = (\mathbf{I} - \mathbf{Q}_\theta)^{-1}$ be the fundamental matrix, and $\mathbf{1}$ be the all-ones vector of dimension $|\mathcal{S}|$. Then:

1. The expected victory time from state $\mathbf{x}$ is

$$\mathbb{E}[\tau_\theta(\mathbf{x})] = [\mathbf{N}_\theta \cdot \mathbf{1}]_{\mathbf{x}}. \quad (23)$$

2. The variance is

$$\text{Var}(\tau_\theta(\mathbf{x})) = [(2\mathbf{N}_\theta - \mathbf{I}) \cdot \mathbf{t}_\theta]_{\mathbf{x}} - [\mathbf{t}_\theta]_{\mathbf{x}}^2 \quad (24)$$

for $\mathbf{t}_\theta = \mathbf{N}_\theta \cdot \mathbf{1}$ the expected hitting time vector.

3. The probability of victory in exactly k turns is

$$\mathbb{P}(\tau_\theta(\mathbf{x}) = k) = [\mathbf{Q}_\theta^{k-1} \cdot \mathbf{r}_\theta]_{\mathbf{x}}. \quad (25)$$

Proof. These are standard results for absorbing Markov chains. The matrix $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$'s role in computing expected absorption times is developed in Kemeny and Snell [3], Chapter 3, Section 3. $\square$

Definition (Initial State). The Cult begins with one garden in a chosen clearing. Without loss of generality, we fix this as Fox, giving $\mathbf{G}_0 = (1, 0, 0)$ and $V_0 = 0$. The Cult also draws 3 cards from the full 54-card deck, so the initial hand state $\mathbf{h}_0$ is random. Under the fresh-deck approximation, the initial distribution is

$$\pi_0(\mathbf{h}) = \prod_{s \in \{F, R, M\}} q(3)^{h_s} (1 - q(3))^{1-h_s} \quad (26)$$

where $q(3) \approx 0.537$ is the probability of drawing at least one card of a given suit in 3 draws (c.f. Proposition 2). The expected victory time from this initial distribution is

$$\mathbb{E}[\tau_\theta] = \sum_{\mathbf{h} \in \{0,1\}^3} \pi_0(\mathbf{h}) \cdot [\mathbf{t}_\theta]_{((1,0,0),0,\mathbf{h})} \quad (27)$$

where $\mathbf{t}_\theta = (\mathbf{I} - \mathbf{Q}_\theta)^{-1} \mathbf{1}$ is the hitting time vector.

The transition matrix $\mathbf{P}_\theta$ is sparse and can be easily constructed; the algorithm is in (Appendix A).

4 OPTIMAL THRESHOLD DERIVATION

We prove that $\theta^* = 2$ is optimal via two approaches: delay value analysis and Markov chain computation.

4.1 Delay Value Characterization

We calculate the expected VP change from waiting an additional turn before scoring a suit.

Definition (Delay Value). For a suit with g gardens, the delay value is

$$\Delta(g) = \mathbb{E}[\text{VP delay-then-score}] - \mathbb{E}[\text{VP score-now}].$$

Immediate scoring is optimal when $\Delta(g) \leq 0$.

Let μ be the growth probability and $p_\ell = 1 - e^{-\lambda/|G|}$ be the per-suit loss probability ($= 1 - e^{-\lambda/3}$ under diversified play). If we delay one turn, the garden count evolves to $g' \in \{g-1, g, g+1\}$: with probability $\mu(1-p_\ell)$ the garden grows and survives, with probability $(1-\mu)(1-p_\ell)$ nothing changes, and with probability p_ℓ a garden is lost (assuming that attrition removes at most one garden per suit, valid for small λ). Then

$$\begin{aligned} \mathbb{E}[V_{\text{delay}}] &= \mu(1-p_\ell)S(g+1) + (1-\mu)(1-p_\ell)S(g) \\ &\quad + p_\ell S(g-1) \\ \implies \Delta(g) &= \mu(1-p_\ell)[S(g+1) - S(g)] \\ &\quad - p_\ell[S(g) - S(g-1)]. \end{aligned}$$

Evaluating at each garden count with parameters $\mu = 0.4$, $p_\ell = 0.065$ (corresp. to $\mu = 0.4$, $\lambda = 0.2$):

g	1	2	3	4	5
$S(g)$	0	2	2	3	4
Gain	2	0	1	1	—
Loss	0	2	0	1	1
$\Delta(g)$	+0.75	-0.13	+0.37	+0.31	-0.065

Two features stand out. At $g = 1$, scoring yields 0 VP, so there is no opportunity cost to waiting; $\Delta(1) > 0$ for all parameter values. At $g = 2$, the scoring plateau $S(2) = S(3) = 2$ means growth provides zero upside, but attrition to $g = 1$ destroys all scoring value (downside = 2). In particular, $\Delta(2) = -2p_\ell < 0 \forall \mu$.

4.2 Optimal Threshold

We show that $\theta^* = 2$ by considering the build-up time. In particular, for $\mu \in [0.2, 0.6]$ and $\lambda \in [0.1, 0.5]$, the optimal threshold is $\theta^* = 2$.

We provide some reasoning for this idea, but not a formal proof.

Claim 5. ($\theta = 1$ is dominated). Since $S(1) = 0$, a policy that scores at $g = 1$ earns zero VP from any suit with exactly one garden. Since $S(g) \geq 2$ for all $g \geq 2$, any threshold $\theta \geq 2$ dominates $\theta = 1$ always.

Claim 6. (Immediate scoring is optimal at $g = 2$). As shown in the table above, $\Delta(2) = -2p_\ell < 0$ for all $\lambda > 0$. This reflects the structure of S : the plateau $S(2) = S(3) = 2$ means growth from $g = 2$ to $g = 3$ provides zero VP, but attrition from $g = 2$ to $g = 1$ causes a total loss of scoring value.

Claim 7. ($\theta \geq 3$ is suboptimal). Although $\Delta(3)$ and $\Delta(4)$ are positive, a threshold policy must first build to that level before scoring begins.

For $\theta = 3$, the case is immediate: since $S(3) = S(2)$ and the draw bonus activates at $g = 2$, the scoring rate is identical to $\theta = 2$'s, and $\theta = 3$ requires strictly more build-up time so $\mathbb{E}[\tau(3)] > \mathbb{E}[\tau(2)] \forall \mu, \lambda$.

For $\theta \geq 4$, we use a decomposition (Appendix C): $\mathbb{E}[\tau(\theta)] \approx T_{\text{build}}(\theta) + 30/\bar{V}(\theta)$, where $\bar{V}(\theta) = S(\theta) \cdot \mathbb{E}[K | n(\theta)] \cdot p_{\text{surv}}$ is the expected VP per scoring turn.

Since all thresholds $\theta \geq 2$ yield $n(\theta) = 4$ (three suits above the draw bonus threshold), $\mathbb{E}[K | n]$ and p_{surv} are common factors. The saving from raising the threshold to θ is therefore

$$\Delta_{\text{score}}(\theta) = \frac{30}{S(2) \cdot C} - \frac{30}{S(\theta) \cdot C} = \frac{30}{C} \cdot \frac{S(\theta) - S(2)}{S(2) \cdot S(\theta)}$$

where $C = \mathbb{E}[K | n=4] \cdot p_{\text{surv}}$ is independent of θ . Since $C \leq 3$ (at most 3 suits scored) and $S(\theta) \leq 4$,

$$\Delta_{\text{score}}(\theta) \leq \frac{30}{C} \cdot \frac{S(\theta) - 2}{2 \cdot S(\theta)} \leq \frac{30}{C} \cdot \frac{1}{4} = \frac{7.5}{C} \quad (28)$$

using $S(5) = 4$, the largest available value. Since $C \geq 1$, this gives $\Delta_{\text{score}} \leq 7.5$ turns saved.

Meanwhile, the additional build-up cost satisfies

$$\Delta_{\text{build}}(\theta) = T_{\text{build}}(\theta) - T_{\text{build}}(2) \geq \frac{\theta - 2}{\mu - \lambda/3} \quad (29)$$

since the net per-suit drift is at most $\mu - \lambda/3$, and all three suits must reach θ . For the parameter range $\mu \in [0.2, 0.6]$ and $\lambda \in [0.1, 0.5]$, the net drift $\mu - \lambda/3$ ranges from 0.03 to 0.57, giving:

θ	$\Delta_{\text{score}} \leq$	$\Delta_{\text{build}} \geq$	Net
4	$\frac{30}{C} \cdot \frac{1}{6} \approx 2.5$	3.5 to 67	loss
5	$\frac{30}{C} \cdot \frac{1}{4} \approx 3.8$	5.3 to 100	loss

(with the most favorable value $C = 2$). Under the most favorable reasonable drift ($\mu = 0.6$, $\lambda = 0.1$), Δ_{build} exceeds Δ_{score} for $\theta \geq 4$, so in the viable regime $\mu > \lambda/3$, the build penalty dominates.

4.3 Endgame Adjustment

When $V + S(G_s) \geq 30$, the optimal action is to and score immediately and win, regardless of θ .

Corollary 8 (Victory-Aware Policy). *The optimal threshold is state-dependent:*

$$\theta^*(V) = \min(\theta_{\text{base}}^*, g^* \text{ s.t. } S(g^*) \geq 30 - V). \quad (30)$$

4.4 Exact Markov Chain Computation

The transition matrix $\mathbf{P}_\theta$ defined in Section 3 enables exact computation of $\mathbb{E}[\tau(\theta)]$.

Remark (Exact Victory Times). Let π_0 be the initial distribution from Definition 3.4, with $\mathbf{G}_0 = (1, 0, 0)$ and $\mathbf{h}_0$ drawn from 3 initial cards. The fundamental matrix $\mathbf{N}_\theta = (\mathbf{I} - \mathbf{Q}_\theta)^{-1}$ yields the hitting time vector $\mathbf{t}_\theta = \mathbf{N}_\theta \cdot \mathbf{1}$, and the expected victory time is $\mathbb{E}[\tau_\theta] = \pi_0 \cdot \mathbf{t}_\theta$ (equation (27)). Solving the sparse linear system $(\mathbf{I} - \mathbf{Q}_\theta)\mathbf{t} = \mathbf{1}$ gives exact values for these quantities.

θ	1	2	3	4	5
$\mathbb{E}[\tau_\theta]$	∞	10.95	12.47	13.85	16.03
$\text{Var}(\tau_\theta)$	—	6.68	9.04	11.31	14.91

Parameters: $\mu = 0.4$, $\lambda = 0.2$. Expectations taken over both the initial hand draw and the trajectory.

5 DIVERSIFICATION ANALYSIS

The threshold policy π_θ prescribes *when* to score, but the Cult must also decide *how to build*: should gardens be spread evenly across suits or concentrated into one? The Markov chain model provides an answer.

Under $\theta^* = 2$, a garden at $g = 1$ contributes nothing, and a garden at $g = 3$ contributes no more than $g = 2$ since $S(3) = S(2)$. The strategy question is therefore whether to activate all three scoring tracks at $g = 2$ or to push a suit higher at the expense of another.

Proposition 9 (Diversification Dominance). *Let $\mathbf{G}_D = (2, 2, 2)$ and $\mathbf{G}_C = (4, 2, 0)$ be two configurations with 6 total gardens. Then*

$$\mathbb{E}[V_{\text{turn}} | \mathbf{G}_D] > \mathbb{E}[V_{\text{turn}} | \mathbf{G}_C]. \quad (31)$$

Proof. (Sketch). The configurations differ in two quantities from our model. First, $\mathbf{G}_D$ activates all three draw bonuses ($n = 4$), while $\mathbf{G}_C$ activates only two ($n = 3$). By Proposition 2, the per-suit hit probability increases from $q(3) = 0.540$ to $q(4) = 0.652$.

Second, $\mathbf{G}_D$ opens three scoring tracks (each worth $S(2) = 2$), while $\mathbf{G}_C$ opens only two (worth $S(4) + S(2) = 3 + 2 = 5$ combined).

Furthermore, at $\lambda = 0.2$, the per-suit survival probability for the diversified strategy is $p_{2,\text{surv}} = e^{-\lambda \frac{2}{6}} \approx 0.936$, the same as the concentrated strategy in its 2-garden suit, but in the 4-garden suit it is $p_{4,\text{surv}} = e^{-\lambda \frac{4}{6}} \approx 0.875$. Therefore:

$$\mathbb{E}[V_{\text{turn}} \mid \mathbf{G}_D] = 3 \times S(2) \times q(4) \times p_{2,\text{surv}} = 3.66 \text{ VP/turn.}$$

$$\begin{aligned} \mathbb{E}[V_{\text{turn}} \mid \mathbf{G}_C] &= [S(4)p_{4,\text{surv}} + S(2)p_{2,\text{surv}}] \times q(3) \\ &= 2.43 \text{ VP/turn.} \end{aligned}$$

Notably, diversification yields a 50% higher rate. $\square$

These results suggest a natural build sequence. In the early game, the Cult should prioritize reaching $(2, 2, 2)$, building one suit at a time: $(2, 0, 0) \rightarrow (2, 2, 0) \rightarrow (2, 2, 2)$, since each new suit at $g = 2$ unlocks a draw bonus and scoring. Further gardens serve as attrition buffers but should not delay scoring: the delay value analysis (Section 4) shows $\Delta(2) < 0$, indicating that the Cult should score any suit at $g \geq 2$ immediately until the endgame adjustment (Section 4.3) overrides the base threshold.

6 MONTE CARLO SIMULATION

We simulate $N = 10,000$ games per configuration via an algorithm (Appendix D, Appendix E).

Table 1: Victory time by threshold (mean), $\mu = 0.4$.

Policy	$\lambda = 0.15$		$\lambda = 0.30$	
	Mean	95% CI	Mean	95% CI
Threshold-2	10.2	± 0.04	11.1	± 0.05
Threshold-3	11.8	± 0.04	13.0	± 0.06
Threshold-4	13.3	± 0.05	14.8	± 0.08

Table 2: Diversification comparison ($\mu = 0.4$, $\lambda = 0.2$, Diversified = $(2, 2, 2)$, Concentrated = $(4, 2, 2)$).

Strategy	Mean T	VP/turn	$\mathbb{P}(\text{win by } t=10)$
Diversified	10.0	3.1	0.67
Concentrated	10.9	2.9	0.48

In the simulation, the Concentrated strategy produces better results than expected (though still worse than Diversified). Inspecting trajectories shows that this is due

to the two strategies often being ‘effectively’ equivalent due to the difficulty of maintaining the $(2, 2, 2)$ position that allows constant full scoring.

Table 3: Sensitivity of $\mathbb{E}[T]$ to parameters ξ (base $\mu = 0.4$, $\lambda = 0.2$, $\theta = 2$).

Parameter	$\partial \mathbb{E}[T] / \partial \xi$	Interpretation
Attrition λ	+5.6	0.1 increase adds 0.56 turns
Growth μ	-17.5	0.1 increase saves 1.75 turns
Threshold θ	+1.7	1 unit adds 1.7 turns

Growth rate is the dominant parameter, and the hardest to model, given that it depends so heavily on a player’s luck with Building and Conspiracies. In real play, higher μ is likely to also yield higher λ , as the higher growth from Sanctify conspiracies is only possible if the Cult’s warriors are frequently attacked. The distribution is right-skewed with occasional long games ($T > 15$) due to poor draws or heavy attrition.

7 CONCLUSION

By developing a Markov chain framework for Lizard Cult scoring in *Root*, we found that $\theta^* = 2$ is robustly optimal because it is simultaneously the minimum viable scoring level and the draw bonus activation threshold. We also saw that the one-turn delay value $\Delta(g)$ is positive at $g = 1$ and negative at $g = 2$, showing that there is risk and no upside to waiting after obtaining 2 gardens in a suit. Ultimately, multi-suit scoring with configuration $(2, 2, 2)$ yields higher VP rates than blind growth.

REFERENCES

- [1] C. Wehrle, *Root: A Game of Woodland Might and Right & The Underworld Expansion*, Leder Games, 2019.
- [2] S. M. Ross, *Introduction to Probability Models*, 12th ed., Academic Press, 2019.
- [3] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, Springer-Verlag, New York, 1976.
- [4] N. L. Johnson, A. W. Kemp, and S. Kotz, *Univariate Discrete Distributions*, 3rd ed., Wiley, 2005.

APPENDIX

Full Appendix (A-F) is accessible through *The Delta Epsilon* online version.